Spectral Shrunk Clustering: Reproduction, Certificates, and When Shrinkage Helps

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Abstract

Spectral Shrunk Clustering (SSC) refines a spectral representation using a convex fidelity–fusion objective. We study its motivation, optimization and experimental evidence, and reconstruct nine methods on eight historical benchmarks. Four benchmarks support measured partial reproduction; other cells retain explicit data, method or subset limitations. We separate a fixed controlled protocol from retrospective numerical matching to the published table. A feasible-dual audit distinguishes optimization error from clustering quality, and label-assisted graph interventions test whether incorrect fusion edges explain weak shrinkage gains. We also analyze an existing truncated fusion penalty as an SSC adaptation, without claiming its novelty or a global certificate for its nonconvex optimization. Results and limitations are generated from auditable scalar trials.

1 Problem, motivation, and the paper's contribution

Spectral clustering represents samples by eigenvectors of a graph Laplacian before applying K-means. This can reveal geometry missed by clustering the original pixels, but the representation and graph are imperfect. Chang et al. [Chang et al., 2015] propose learning *shrunk patterns* from the spectral coordinates: nearby embedded samples should move together while retaining fidelity to their initial positions. Their contribution is a representation-refinement formulation, removal of an orthogonality constraint from this refinement, and an iterative optimization procedure with empirical clustering evidence. SSC here means *Spectral Shrunk Clustering*, not Sparse Subspace Clustering.

Let A be the input-space affinity and D_A its degree matrix. The eigenvectors for the c smallest eigenvalues of $L_A = I - D_A^{-1/2} A D_A^{-1/2}$ form $F \in \mathbb{R}^{n \times c}$. Unlike PCA's variance criterion, this embedding depends on neighborhood geometry. Unlike LDA, labels do not enter our embedding or shrinkage functions; the class count follows the benchmark protocol.

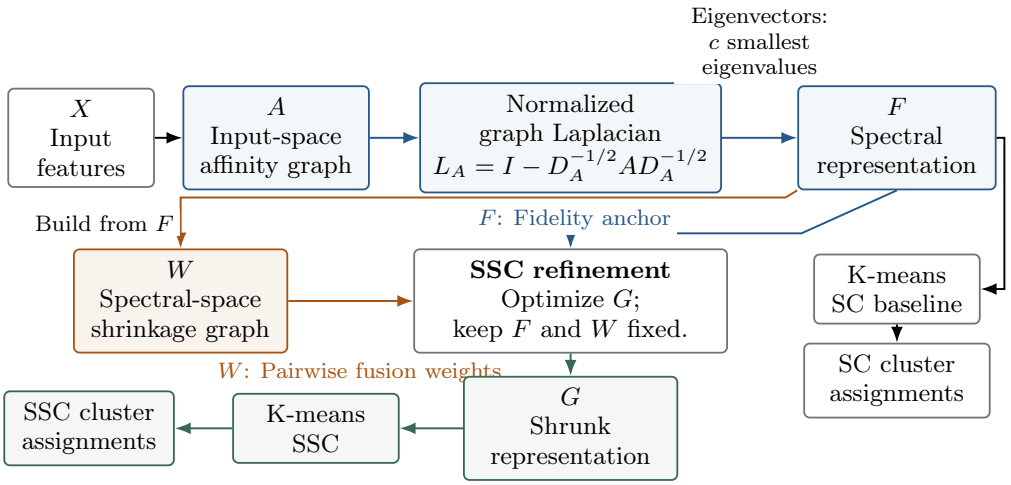
2 Method and what convexity guarantees

Figure 1 separates the two graph roles: A determines the initial spectral geometry, while W, built from F, sets pairwise fusion strengths. F remains the fidelity anchor. Changing W with F fixed changes refinement; changing A generally also changes F and hence W. With an oriented incidence matrix B storing each undirected edge once, Equation 8 is

$$P_0(G) = \sum_i \|g_i - f_i\|_2 + 2\gamma \sum_e w_e \|(BG)_e\|_2. \tag{1}$$

Both terms are convex for fixed F and nonnegative W, but uniqueness is not guaranteed. The authors use iteratively reweighted quadratic problems. Under the ordered-pair convention, the stationary

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SSC refinement objective (Equation 8)

$$\min_G \Big[\underbrace{\sum_i \|g_i - f_i\|_2}_{\text{Fidelity to } F} + \underbrace{2\gamma \sum_{i<j} W_{ij} \|g_i - g_j\|_2}_{\text{Fusion controlled by } W}\Big]$$

Each undirected edge once; both norms unsquared.

Figure 1: Two graphs with different roles in spectral shrunk clustering. The input-space affinity A defines the Laplacian used to obtain the spectral representation F. A second graph W, constructed in the spectral space, controls pairwise fusion during refinement. SSC optimizes G while keeping F and W fixed, then applies K-means to G. The SC baseline applies K-means directly to the same F.

system has $S + 2\gamma\widetilde{L}$ (E8); the printed Equation 13 has $S + \gamma\widetilde{L}$ (E13). E13 at nominal gamma therefore targets E8 at effective gamma/2. Comparing these unequal objectives at the same nominal gamma would create a false solver discrepancy.

Convexity concerns G with F/W fixed. Graph construction, selecting an eigenspace and final K-means remain separate stages. A global representation optimum need not give globally optimal labels or higher ACC. Signs and orthogonal rotations preserve Euclidean geometry; changes to the selected eigenspace at a spectral cutoff can matter. Row normalization is a preprocessing change, not an eigenpair-residual test.

3 Published evidence and our partial reproduction

The paper compares nine methods on AR, JAFFE, ORL, UMIST, BinAlpha, MSRA50, YaleB and USPS, reports 50 clustering repetitions, and describes a parameter grid $\{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$. SSC has the highest reported ACC across these benchmarks. This does not extend to every metric: YaleB SSC ACC is 49.8 versus SC 34.8, while SSC NMI is 46.8 versus SC 55.6 (*Reported in Chang et al. (2015)*). ACC uses optimal label matching; our NMI uses geometric entropy normalization. The paper also describes selecting randomized runs by objective. A selected assignment and a per-run mean/SD are distinct endpoints, so we retain both rather than attaching 50-run SD to a single assignment.

Our source-based implementations include one-shot SLPS (called SPLS in the SSC table), following Hou et al. [Hou et al., 2013]. LDA-SC remains blocked because its named algorithm is insufficiently specified. AR's required subset, historical UMIST564 and MSRA50 remain data-blocked. BinAlpha subset identity is unverified; UMIST575 and candidate BinAlpha results are supplementary, not replacements. The historical matrix has 32 completed, 24 data-blocked, 8 method-blocked and 8 subset-unverified cells.

Table 1: Reported in Chang et al. (2015) versus our measured ACC (percent). Closest is retrospective ACC matching. Fixed is Y0/E8/gamma=1/random. Full deviations are described in the text and appendix.

Data	Method	Paper	Closest mean ± SD	Fixed mean ± SD	Mark
ORL	SC	72.7	72.725 ± 3.149	56.180 ± 4.193	V
ORL	SSC	76.0	75.965 ± 1.367	57.135 ± 3.955	V,T
JAFFE	SC	76.1	68.526 ± 11.152	68.526 ± 11.152	R
JAFFE	SSC	81.2	74.751 ± 8.371	68.432 ± 11.081	R,T
YaleB	SC	34.8	34.362 ± 0.957	18.340 ± 1.284	Y3,G
YaleB	SSC	49.8	43.583 ± 1.287	17.629 ± 1.190	Y3,G,I
USPS	SC	64.3	65.597 ± 1.867	61.739 ± 8.086	G
USPS	SSC	75.5	76.844 ± 5.117	61.698 ± 8.089	V,I

R: random; V: vanilla++; G: sklearn greedy++; T: 300-step returned iterate, optimum not certified; I: historical iterate criterion, no dual certificate.

Table 2: New seeded Y0 representations, shared within each SC/SSC pair; greedy++ seeds 0–49. ACC percentages. U: uncertified output. Best certified is not the complete oracle if other configurations remain uncertified.

Data	SC	SSC gamma=1	Best certified in Γ_5	Coverage
ORL	78.33	78.39	78.39	5/5 certified
JAFFE	95.31	95.10	95.31	5/5 certified
YaleB	21.78	21.55	21.78	5/5 certified
USPS	65.60	65.40 U	65.60	2/5 certified

Two distinct questions. Closest is *Our adapted setting*: retrospective absolute matching of the unrounded ACC mean to the paper, using every planned batch-0 seed. NMI/SD follow the same configuration and trials. This neither recovers the author protocol nor selects without labels. Additional batches assess repeatability. Fixed is *Our reproduction*: Y0/E8/gamma=1/random, one Lloyd start per seed. PCA/SLPS fixed cells remain uncollapsed because no unique reference was locked.

The matched values need their settings. ORL's closest SSC uses E13, nominal gamma=1000 and vanilla++; JAFFE uses E13, 10^6 and random. Both are 300-step truncated outputs. YaleB's closest SSC uses DengCai32, Y3 and greedy++; Y3 changes both input L2 and embedding row normalization. USPS uses E8, gamma=1000 and vanilla++. An iterate stopping criterion does not certify optimality. Full settings, NMI and all 72 cells are in the appendix and generated manifests.

4 Critical analysis and performance-driven extensions

Optimization versus the target. We implement standard PDHG [Chambolle and Pock, 2011] and recover feasible dual variables instead of using agreement between two solvers as a certificate. For $c_e = 2\gamma w_e$, any q satisfying $\|q_e\| \le c_e$ and $\|(B^T q)_i\| \le 1$ gives $D(q) = \langle B^T q, F\rangle \le \min P_0$. We retain actual best primal/dual points and require relative gap $\le 10^{-5}$. Label-free IRLS initialization can tighten the primal bound; its acceptance uses the same objective, not ACC. Historical outputs and fresh seeded representations remain distinct artifacts.

At the best certified original gamma in the five-value grid, the prespecified incorrect-edge gate supports 0/4 benchmarks; two are required. No real-data capped-fusion improvement or achievement of the 2-point target is established. Uncertified controls make some conclusions inconclusive, rather than disproving the mechanism. Analytic and synthetic checks are reported separately.

Why incorrect fusion is a hypothesis. An edge in W attracts its endpoints even if their true classes differ. We separately measure A/W pollution, false merges, false splits and within/between-class contraction. Oracle deletion of cross-class W edges uses labels only for diagnosis; random deletion, equal remaining edge mass and static length filtering distinguish targeted correction from simply weakening shrinkage. At the best certified original gamma, a prespecified gate requires certified gains over these controls on at least two benchmarks, with nondecreasing NMI. Length filtering additionally needs evidence that erroneous edges are distinguishable by length. Failure or uncertainty at this gate does not establish a universal SSC failure.

YaleB cropped W has 63.2% cross-class edges/70.2% mass. Length AUROC 0.380 means wrong edges tend to be shorter, challenging long-edge truncation. These are W/F-space diagnostics.

Existing robust fusion, adapted carefully. We examine replacing edge distance r by $\min(r, \tau)$, an existing group-truncated penalty [Pan et al., 2013, Wu et al., 2016]. RCC similarly motivates weakening spurious pairwise relations [Shah and Koltun, 2017]; this is different from robustifying anchor fidelity. For two group-constant representatives with sizes $m < n$ and separation $d > \tau$, the original optimum is $d\min(m, \lambda)$, versus $\min(md, \lambda\tau)$ for capped fusion. Thus $m < \lambda < md/\tau$ permits original fusion while capping preserves separation. This proves a narrow information-preservation mechanism, not better real-data ACC. Squared/non-squared fidelity crossed with linear/capped fusion separates anchor and edge effects. Nonconvex active-edge updates receive inner convex certificates only; initialization dependence and static-screening controls remain necessary.

Evaluation discipline. The practical target is four-benchmark mean ACC gain of at least 2 points with nondecreasing NMI. A 15-configuration capability oracle and locked no-label operating point are distinct; neither complete comparison is claimed when C is not triggered. Shared K-means seeds give conditional rounding intervals, not independent dataset replicates or unbiased post-selection coverage. Huber activation and heat stability alone do not establish label improvement.

The historical USPS gamma=1000 greedy++ point has ACC 79.918% and NMI 85.609%. Fresh uncertified poor outputs on separately regenerated F/W do not invalidate it. Synthetic bridge tests show damage avoidance shared by static deletion and squared fidelity, with SC already perfect. Geometry stability and Huber MSE changes alone do not establish label gains.

5 Conclusion and limitations

SSC makes refinement interpretable, but label quality depends on geometry, edges, optimization and rounding. Our contribution is traceable partial reproduction and conditional mechanism analysis. Incomplete certification, historical array identity, within-dataset development limit conclusions. Code, report sources and negative findings are at GitHub v3.

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A Historical cells, selection, and effective settings

Paper values are *Reported in Chang et al. (2015)*; measured columns are *Our reproduction/adapted setting*. Units are percent. Closest uses all 50 prescribed batch-0 seeds. NMI follows the same selected configuration; other batches and selected min-inertia assignments remain distinct endpoints.

A.1 Closest display: 72 historical cells

Data	Method	Paper ACC	Measured ACC	Measured NMI	Status / setting
AR	kmeans	36.3	–	–	data blocked
AR	sc	41.6	–	–	data blocked
AR	pca_kmeans	39.8	–	–	data blocked
AR	pca_sc	43.2	–	–	data blocked
AR	lda_kmeans	40.4	–	–	data blocked
AR	lda_sc	44.5	–	–	method blocked
AR	llc	48.7	–	–	data blocked
AR	spls	49.2	–	–	data blocked
AR	ssc	51.3	–	–	data blocked
JAFFE	kmeans	75.6	73.192 ± 5.734	83.068 ± 3.452	measured; Y0, R
JAFFE	sc	76.1	68.526 ± 11.152	83.016 ± 5.724	measured; Y0, R
JAFFE	pca_kmeans	75.8	74.826 ± 7.714	83.106 ± 4.531	measured; Y0, R
JAFFE	pca_sc	76.9	80.028 ± 5.737	87.786 ± 2.965	measured; Y0, V
JAFFE	lda_kmeans	76.5	64.441 ± 9.187	78.602 ± 5.082	measured; Y0, full refit
JAFFE	lda_sc	77.4	–	–	method blocked
JAFFE	llc	78.6	91.549 ± 0.000	92.745 ± 0.000	measured; Y0, Yu-Shi
JAFFE	spls	79.5	80.854 ± 6.576	86.915 ± 3.582	measured; Y0, one_shot
JAFFE	ssc	81.2	74.751 ± 8.371	85.600 ± 4.325	truncated; Y0, E13, g=1e+06, R
ORL	kmeans	60.5	61.455 ± 2.851	82.755 ± 1.166	measured; Y0, R
ORL	sc	72.7	72.725 ± 3.149	87.492 ± 1.223	measured; Y0, V
ORL	pca_kmeans	64.5	63.115 ± 3.275	83.585 ± 1.224	measured; Y0, R
ORL	pca_sc	67.8	67.570 ± 2.925	85.810 ± 1.274	measured; Y0, V
ORL	lda_kmeans	65.6	56.305 ± 4.142	81.003 ± 1.652	measured; Y0, full refit
ORL	lda_sc	68.3	–	–	method blocked
ORL	llc	71.5	81.070 ± 0.606	89.917 ± 0.270	measured; Y0, Yu-Shi
ORL	spls	74.2	74.195 ± 1.982	87.898 ± 0.866	measured; Y0, one_shot
ORL	ssc	76.0	75.965 ± 1.367	90.008 ± 0.411	truncated; Y0, E13, g=1000, V
UMIST	kmeans	41.3	–	–	data blocked
UMIST	sc	52.2	–	–	data blocked
UMIST	pca_kmeans	48.8	–	–	data blocked
UMIST	pca_sc	54.1	–	–	data blocked
UMIST	lda_kmeans	49.7	–	–	data blocked
UMIST	lda_sc	54.7	–	–	method blocked
UMIST	llc	63.3	–	–	data blocked
UMIST	spls	70.4	–	–	data blocked
UMIST	ssc	71.1	–	–	data blocked
BinAlpha	kmeans	41.7	–	–	subset unverified
BinAlpha	sc	43.6	–	–	subset unverified
BinAlpha	pca_kmeans	42.4	–	–	subset unverified
BinAlpha	pca_sc	44.3	–	–	subset unverified
BinAlpha	lda_kmeans	42.9	–	–	subset unverified
BinAlpha	lda_sc	45.1	–	–	method blocked
BinAlpha	llc	40.7	–	–	subset unverified
BinAlpha	spls	48.5	–	–	subset unverified
BinAlpha	ssc	49.4	–	–	subset unverified
MSRA50	kmeans	46.2	–	–	data blocked
MSRA50	sc	52.3	–	–	data blocked
MSRA50	pca_kmeans	56.0	–	–	data blocked
MSRA50	pca_sc	54.9	–	–	data blocked
MSRA50	lda_kmeans	56.4	–	–	data blocked

Data	Method	Paper ACC	Measured ACC	Measured NMI	Status / setting
MSRA50	lda_sc	55.1	–	–	method blocked
MSRA50	llc	48.1	–	–	data blocked
MSRA50	spls	60.3	–	–	data blocked
MSRA50	ssc	63.2	–	–	data blocked
YaleB	kmeans	14.4	9.741 ± 0.465	13.250 ± 0.585	measured; Y0, G; DengCai32
YaleB	sc	34.8	34.362 ± 0.957	45.145 ± 0.731	measured; Y3, G
YaleB	pca_kmeans	24.9	9.848 ± 0.553	13.317 ± 0.662	measured; Y0, G; DengCai32
YaleB	pca_sc	36.8	30.925 ± 0.543	38.859 ± 0.951	measured; Y1, G; DengCai32
YaleB	lda_kmeans	26.1	9.799 ± 0.847	13.786 ± 1.145	measured; Y0, full refit; DengCai32
YaleB	lda_sc	38.2	–	–	method blocked
YaleB	llc	38.2	38.241 ± 0.069	45.232 ± 0.093	measured; Y0, Yu-Shi; DengCai32
YaleB	spls	47.3	13.522 ± 0.776	19.438 ± 0.689	measured; Y0, one_shot; DengCai32
YaleB	ssc	49.8	43.583 ± 1.287	54.906 ± 0.826	measured; Y3, E8, g=0.001, G; DengCai32
USPS	kmeans	65.4	66.059 ± 2.175	61.042 ± 0.874	measured; Y0, R
USPS	sc	64.3	65.597 ± 1.867	80.315 ± 2.128	measured; Y0, G
USPS	pca_kmeans	69.4	66.288 ± 1.948	60.991 ± 0.668	measured; Y0, R
USPS	pca_sc	69.1	65.881 ± 2.278	80.761 ± 1.029	measured; Y0, G
USPS	lda_kmeans	70.1	58.613 ± 6.332	60.292 ± 4.215	measured; Y0, full refit
USPS	lda_sc	70.4	–	–	method blocked
USPS	llc	63.9	63.108 ± 0.005	73.987 ± 0.002	measured; Y0, Yu-Shi
USPS	spls	71.4	71.358 ± 4.134	73.422 ± 1.411	measured; Y0, one_shot
USPS	ssc	75.5	76.844 ± 5.117	84.082 ± 2.957	measured; Y0, E8, g=1000, V

A.2 Fixed protocol: 72 cells, with measured grids retained

Data	Method	Paper ACC	Measured ACC	Measured NMI	Status / setting
AR	kmeans	36.3	–	–	data blocked
AR	sc	41.6	–	–	data blocked
AR	pca_kmeans	39.8	–	–	data blocked
AR	pca_sc	43.2	–	–	data blocked
AR	lda_kmeans	40.4	–	–	data blocked
AR	lda_sc	44.5	–	–	method blocked
AR	llc	48.7	–	–	data blocked
AR	spls	49.2	–	–	data blocked
AR	ssc	51.3	–	–	data blocked
JAFFE	kmeans	75.6	73.192 ± 5.734	83.068 ± 3.452	measured; Y0, R
JAFFE	sc	76.1	68.526 ± 11.152	83.016 ± 5.724	measured; Y0, R
JAFFE	pca_kmeans	75.8	–	–	grid not collapsed
JAFFE	pca_sc	76.9	–	–	grid not collapsed
JAFFE	lda_kmeans	76.5	64.441 ± 9.187	78.602 ± 5.082	measured; Y0, full refit
JAFFE	lda_sc	77.4	–	–	method blocked
JAFFE	llc	78.6	94.685 ± 0.743	93.687 ± 0.365	measured; Y0, Yu-Shi
JAFFE	spls	79.5	–	–	grid not collapsed
JAFFE	ssc	81.2	68.432 ± 11.081	83.049 ± 5.721	truncated; Y0, E8, g=1, R
ORL	kmeans	60.5	61.455 ± 2.851	82.755 ± 1.166	measured; Y0, R
ORL	sc	72.7	56.180 ± 4.193	80.471 ± 2.291	measured; Y0, R
ORL	pca_kmeans	64.5	–	–	grid not collapsed
ORL	pca_sc	67.8	–	–	grid not collapsed
ORL	lda_kmeans	65.6	56.305 ± 4.142	81.003 ± 1.652	measured; Y0, full refit

Data	Method	Paper ACC	Measured ACC	Measured NMI	Status / setting
ORL	lda_sc	68.3	–	–	method blocked
ORL	llc	71.5	81.560 ± 0.705	89.993 ± 0.181	measured; Y0, Yu-Shi
ORL	spls	74.2	–	–	grid not collapsed
ORL	ssc	76.0	57.135 ± 3.955	81.165 ± 2.204	measured; Y0, E8, g=1, R
UMIST	kmeans	41.3	–	–	data blocked
UMIST	sc	52.2	–	–	data blocked
UMIST	pca_kmeans	48.8	–	–	data blocked
UMIST	pca_sc	54.1	–	–	data blocked
UMIST	lda_kmeans	49.7	–	–	data blocked
UMIST	lda_sc	54.7	–	–	method blocked
UMIST	llc	63.3	–	–	data blocked
UMIST	spls	70.4	–	–	data blocked
UMIST	ssc	71.1	–	–	data blocked
BinAlpha	kmeans	41.7	–	–	subset unverified
BinAlpha	sc	43.6	–	–	subset unverified
BinAlpha	pca_kmeans	42.4	–	–	subset unverified
BinAlpha	pca_sc	44.3	–	–	subset unverified
BinAlpha	lda_kmeans	42.9	–	–	subset unverified
BinAlpha	lda_sc	45.1	–	–	method blocked
BinAlpha	llc	40.7	–	–	subset unverified
BinAlpha	spls	48.5	–	–	subset unverified
BinAlpha	ssc	49.4	–	–	subset unverified
MSRA50	kmeans	46.2	–	–	data blocked
MSRA50	sc	52.3	–	–	data blocked
MSRA50	pca_kmeans	56.0	–	–	data blocked
MSRA50	pca_sc	54.9	–	–	data blocked
MSRA50	lda_kmeans	56.4	–	–	data blocked
MSRA50	lda_sc	55.1	–	–	method blocked
MSRA50	llc	48.1	–	–	data blocked
MSRA50	spls	60.3	–	–	data blocked
MSRA50	ssc	63.2	–	–	data blocked
YaleB	kmeans	14.4	9.199 ± 0.580	12.413 ± 0.772	measured; Y0, R
YaleB	sc	34.8	18.340 ± 1.284	26.903 ± 0.672	measured; Y0, R
YaleB	pca_kmeans	24.9	–	–	grid not collapsed
YaleB	pca_sc	36.8	–	–	grid not collapsed
YaleB	lda_kmeans	26.1	9.672 ± 0.823	13.416 ± 1.065	measured; Y0, full refit
YaleB	lda_sc	38.2	–	–	method blocked
YaleB	llc	38.2	28.037 ± 0.127	35.369 ± 0.393	measured; Y0, Yu-Shi
YaleB	spls	47.3	–	–	grid not collapsed
YaleB	ssc	49.8	17.629 ± 1.190	26.184 ± 0.900	measured; Y0, E8, g=1, R
USPS	kmeans	65.4	66.059 ± 2.175	61.042 ± 0.874	measured; Y0, R
USPS	sc	64.3	61.739 ± 8.086	74.940 ± 4.494	measured; Y0, R
USPS	pca_kmeans	69.4	–	–	grid not collapsed
USPS	pca_sc	69.1	–	–	grid not collapsed
USPS	lda_kmeans	70.1	58.613 ± 6.332	60.292 ± 4.215	measured; Y0, full refit
USPS	lda_sc	70.4	–	–	method blocked
USPS	llc	63.9	63.108 ± 0.005	73.987 ± 0.002	measured; Y0, Yu-Shi
USPS	spls	71.4	–	–	grid not collapsed
USPS	ssc	75.5	61.698 ± 8.089	75.229 ± 4.558	measured; Y0, E8, g=1, R

A.3 Selection and source trace

The selection manifest contains complete effective parameters, exact initializer names, seeds, batch counts, metadata/source hashes, blocked reasons, execution and solver statuses, and no-data reconstruction commands. PCA/SLPS grids have no unique locked reference. LLC uses its pre-existing ridge=1, bandwidth multiplier=1 reference and Yu–Shi rounding; LDA-Kmeans refits per seed. YaleB cropped is the fixed primary source, DengCai32 is alternate-source sensitivity. UMIST575 and candidate BinAlpha do not replace historical subsets. The immutable scalar package contains 378550

trials, 3059 jobs and 6563 configuration rows. It reconstructs tables without images, old worktrees, Git history or caches. It does not recover unavailable historical representation arrays. The gzip file and decompressed CSV have different hashes. See the machine-readable release manifest.

B Certificates, mechanisms, provenance and resource limits

B.1 Certificate definitions

For radial Huber, $D_\delta(q) = \langle B^T q, F\rangle - (\delta/2)\sum_i \|(B^T q)_i\|^2/a_i$. Edge and node balls apply; squared fidelity removes node balls. An inward capacity-ratio scaling recovers feasible q. Best primal means an actual stored G; objective, array and clustering metrics are never mixed across iterates. Negative primal-dual gaps beyond tolerance are errors. The simple $G = F$ sufficient region uses $b = B^T(2wu)$, with unit directions on noncoincident F-edges and zero otherwise, and $\gamma_0 = 1/\max_i \|b_i\|$. Coincident edges need not yield the maximal onset. Infinity is asserted only for zero-TV analytic cases, not a finite search reaching a large gamma. All 88 historical verified claims were audited: 'ANALYTIC_G_EQUALS_F': 45, 'RECORDED_OBJECTIVE_GAP_CERTIFIED': 37, 'LEGACY_RETURNED_ITERATE_UNCERTIFIED': 6. Direct analytic G=F checks and certification using a recorded scalar primal are distinguished; the latter is not recovery of an unavailable historical G array. Every old status remains in the audit CSV. The numerical amendment retains cold runs and adds an exact-objective-accepted IRLS primal initializer (at most 300 iterations for n<=3000, otherwise 30; at most 60 seconds). Failure falls back to the cold start. It changes neither target, graph, parameter grid nor certificate threshold. The code/config amendment is explicit; development data were already inspected, so later results are not independent held-out validation. Capped fusion uses active-edge block minimization and two initializations. Inner convex gap certification is not a global certificate for the nonconvex objective. Fixed-point, truncation, monotonicity rejection and missing second-start states remain explicit. Tiny-graph state enumeration supplies an independent global reference only for those tiny problems.

B.2 Candidate mechanisms

Radial Huber is convex and orthogonally invariant. Along $G(t) = (1-t)F + t1m^T$, graph TV decreases linearly while near-zero fidelity increases quadratically, so nontrivial F is not optimal for positive gamma/delta. This proves activation, not better ACC. The bound $0 \le r - \phi_\delta(r) \le \delta/2$ gives original-objective error at most $n\delta/2$ plus a certified absolute solver gap. Bounded radial fidelity influence is not a transferred breakdown theorem [Sun et al., 2025]. Heat features use $H_t = e^{-tL/2}$ with $H_t H_t^T = K_t = e^{-tL}$; rows of K_t correspond instead to time 2t distances. Duhamel bounds kernel operator change by $t\|L - L'\|_2$ and each squared-distance change by $2t\|L - L'\|_2$. A kNN boundary gap above $4t\|L - L'\|_2$ cannot cross under that perturbation. Full-spectrum random projection is approximate and does not automatically inherit this exact bound. Neither stability nor a large diffusion time guarantees labels [Coifman and Lafon, 2006, Gasteiger et al., 2019].

B.3 Evaluation and resource limits

The old robustness table did not rerun SSC, and used the wrong USPS pixel domain. It is retained with an invalidity annotation. Corrected perturbations declare [-1,1] for USPS, preserve zero-change identity and rerun the full corresponding pipeline. ARI/Jaccard alone do not measure class accuracy. All primary K-means endpoints use 50 seeds, one greedy++ initialization and sample SD. Paired bootstrap uses seed 20261108 and 2000 resamples. Perturbation seed, not each correlated rounding, is an independent unit. Missing four-benchmark coverage leaves the formal average NA; the two YaleB sources are not independent benchmarks. New computation is capped at 36 CPU-core hours, two workers and a target 8 GiB total RSS, with no GPU or paid resources. PDHG has at most 30000 iterations per fit and 10/30-minute small/large timeouts. Runtime, memory and all retained solver versions are in the resource ledger. Failure never becomes zero ACC. Native prclust 1.3 was compiled at a pinned source revision and executed on a five-point sanity check; this is not a benchmark result. New optimization modules are from-scratch formula implementations. Scientific snapshots, data restrictions, exact source revisions and clone verification are indexed in docs/final. The squared control uses $\sum_i \|g_i - f_i\|^2/(2s_F)$. Multiplying by $2s_F$ maps its linear-fusion coefficient to ordered-pair Equation 7 as $\gamma_{E7} = 2s_F\gamma$. Equal numerical gamma values across fidelities do not

imply equal regularization strength; separate searches are required. For N1 quadratic scale delta, replace s_F by delta. Measured fit/audit CPU is 2.623 core-hours; the conservative two-worker session envelope is 3.822, below 36. Largest recorded per-fit peak RSS is 597.7 MiB. Aggregate RSS was spot-checked, not continuously enforced.

B.4 Separate A and W audit

Label-assisted fractions and AUROC on the newly generated Y0 representations. W length is measured in F, with cross-class edges positive; values below .5 mean longer edges do not identify errors. A native-input length AUROC is not asserted.

Dataset	Graph	cross edges (%)	cross mass (%)	length AUROC
ORL_att92	A	22.31	11.99	–
ORL_att92	W	17.01	4.65	0.830
JAFFE	A	3.55	0.21	–
JAFFE	W	6.87	7.13	0.556
YaleB_cropped2414	A	58.88	53.60	–
YaleB_cropped2414	W	63.24	70.23	0.380
USPS	A	5.04	2.32	–
USPS	W	5.17	2.00	0.758
YaleB_dengcai32	A	51.95	46.64	–
YaleB_dengcai32	W	54.28	63.43	0.360

B.5 Reduced-model derivation and equality cases

For group-constant representatives, let $r = \|u - v\|$. Since $n > m$, the fidelity is at least $m \max(d - r, 0)$, attained on the anchor segment with $v = b$ for $0 \le r \le d$. For linear fusion the reduced cost is $md + (\lambda - m)r$, hence $d \min(m, \lambda)$. At $\lambda = m$ every segment location is optimal. For $d > \tau$, the capped cost is piecewise linear, and its endpoint minimum is $\min(md, \lambda\tau)$; the interior kink cannot improve it. At $md = \lambda\tau$ both fused and separated endpoint solutions exist. Strict inequalities are therefore needed to claim preservation. Graph weights are fixed throughout this statement; rebuilding W as d changes can change lambda. This is not a theorem for arbitrary five-point graphs or classification risk.

C Measured secondary experiments and negative findings

C.1 Synthetic mechanism checks

Synthetic seeds 100–109 are independent instances; each has 50 conditional K-means seeds. Parameters and distance scales are frozen from the unpolluted instance. The full 2-by-2 and static-screening controls are retained, including uncertified inner solves. This intentionally easy geometry can show damage avoidance but is weak evidence of added clustering capacity.

Scenario	Method	ACC	NMI	qualified / 10
anchor_0.2	SC	100.000	100.000	10
anchor_0.2	SSC	100.000	100.000	10
anchor_0.2	radial_capped	100.000	100.000	10
anchor_0.2	squared_linear	100.000	100.000	10
anchor_0.2	static_screen	100.000	100.000	10
bridge_0.1	SC	100.000	100.000	10
bridge_0.1	SSC	99.889	99.555	10
bridge_0.1	radial_capped	100.000	100.000	10
bridge_0.1	squared_linear	100.000	100.000	10
bridge_0.1	static_screen	100.000	100.000	10
imbalanced_bridge	SC	100.000	100.000	10
imbalanced_bridge	SSC	99.156	96.725	10
imbalanced_bridge	radial_capped	100.000	100.000	10
imbalanced_bridge	squared_linear	100.000	100.000	10
imbalanced_bridge	static_screen	100.000	100.000	10

Bridge contamination slightly lowers SSC ACC/NMI; capping, static deletion and squared fidelity all avoid this damage, while SC is already perfect. Capping is therefore neither necessary nor shown to improve beyond SC. At 20% anchor contamination all methods have 100% ACC/NMI, yet capping can worsen representation MSE by removing useful correction. The radial-Huber primary pilot changes MSE slightly without label improvement; its full real-data search is not promoted.

C.2 Tiny global reference and initialization dependence

For the documented five-point graph, 1024 active states are enumerated, 960 excluded by a nonnegative constant lower bound, and 64 convex states independently certified. Original primal/dual are approximately 8; capped global primal/dual are 3. The local capped algorithm returns objective 3 (ACC/NMI 100%) from F, but objective 8 (ACC 60%, NMI 0) from original SSC. Global certification comes solely from enumeration on this tiny graph.

C.3 Secondary heat ablation

The 80-row geometry pilot shows higher projected-heat kNN stability, not label improvement. Primary t=4, r=128 uses the same A, full-spectrum action, fixed projection seed, hard-dimension and W-only controls. Below, each cell is ACC/NMI (percent); common gamma=1 and own $2\gamma_0$ are locked without labels. C/U denote certified/uncertified in that order. Four original gamma=1 controls are reused from Table 2; 28 new fits and their actual gamma/costs are in the scalar release. N2 changes F and cannot substitute for the same-original-F main goal.

Data	Feature	SC ACC/NMI	gamma=1 ACC/NMI	own $2\gamma_0$ ACC/NMI	status
ORL	original	78.33/89.87	78.39/89.74	78.40/89.59	C/C
ORL	hard128	47.97/67.93	44.39/71.72	47.82/67.91	C/C
ORL	heat128	75.19/87.96	77.38/89.07	76.45/88.68	C/C
ORL	W-only	78.33/89.87	77.69/90.05	77.97/89.66	C/C
JAFFE	original	95.31/94.13	95.10/94.06	95.31/94.13	C/C
JAFFE	hard128	23.65/29.29	10.80/0.00	22.59/27.09	C/C
JAFFE	heat128	51.55/60.41	55.42/64.90	53.62/62.34	C/C
JAFFE	W-only	95.31/94.13	95.21/94.23	95.31/94.14	C/C
YaleB	original	21.78/27.73	21.55/27.49	21.79/27.70	C/C
YaleB	hard128	15.81/24.15	14.64/24.89	15.97/24.51	C/C
YaleB	heat128	19.75/24.07	17.05/29.46	19.42/24.12	C/C
YaleB	W-only	21.78/27.73	21.87/27.64	21.83/27.69	C/C
USPS	original	65.60/80.31	65.40/80.11	65.55/80.42	U/U
USPS	hard128	30.32/33.81	26.84/27.72	29.95/33.32	U/C
USPS	heat128	17.23/3.44	17.38/4.47	17.39/3.55	U/C
USPS	W-only	65.60/80.31	63.71/76.77	65.41/80.55	U/C

The exact kernel bound applies before projection. Fixed-subset exact-kernel errors and restricted-candidate kNN recall are separate diagnostics, not full-graph guarantees. No t/r/seed tuning or projection sensitivity is claimed.

C.4 Corrected full-pipeline image conditions

Four fixed image conditions reconstruct A/F/W/SSC at clean gamma=1; USPS uses [-1,1] with full-span brightness steps. Ranges below are descriptive changes in percentage points against each clean method. Conditions are not independent random replicates; no inferential or adaptive-selection claim is made. USPS SSC remains uncertified.

Data	SC ΔACC	SSC ΔACC	SC ΔNMI	SSC ΔNMI
ORL	[-0.06,1.35]	[0.00,1.76]	[-0.23,0.37]	[0.00,0.84]
JAFFE	[0.00,0.00]	[0.00,0.21]	[0.00,0.00]	[0.00,0.08]
YaleB	[0.08,0.18]	[0.03,0.61]	[0.08,0.20]	[0.03,0.42]
USPS	[-0.36,0.09]	[-0.05,0.36]	[-0.32,0.48]	[0.05,0.75]

D Parameter curves and controlled edge interventions

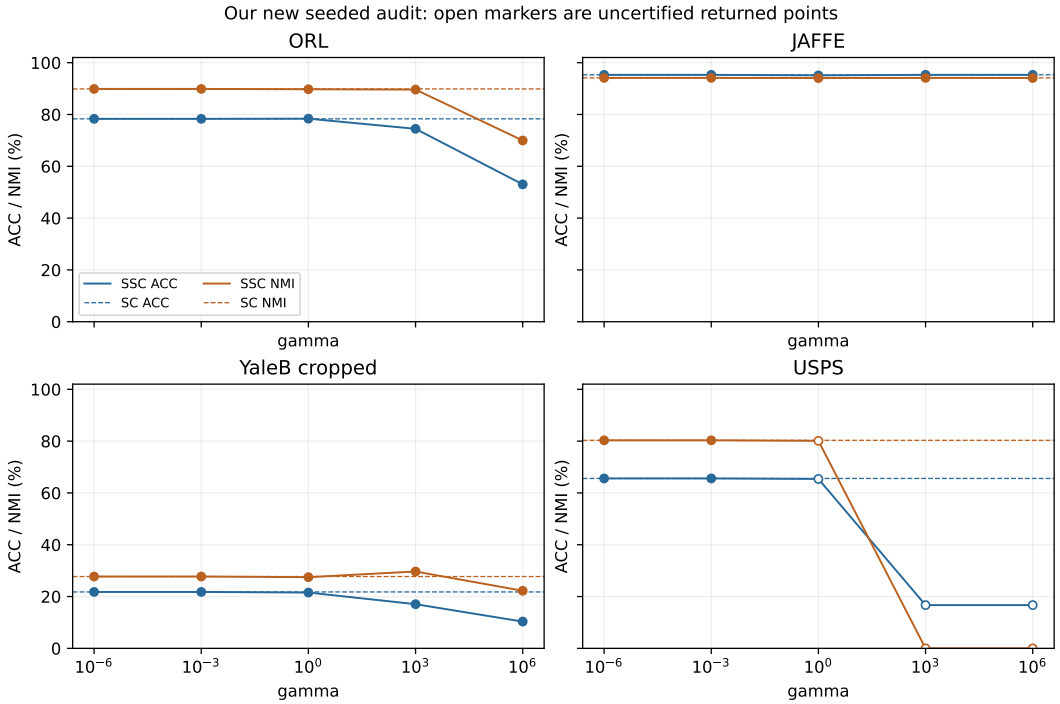


Figure 2: Same seeded features and protocol; open markers are uncertified. These fresh points do not replace stronger historical points on unrecoverable old representations.

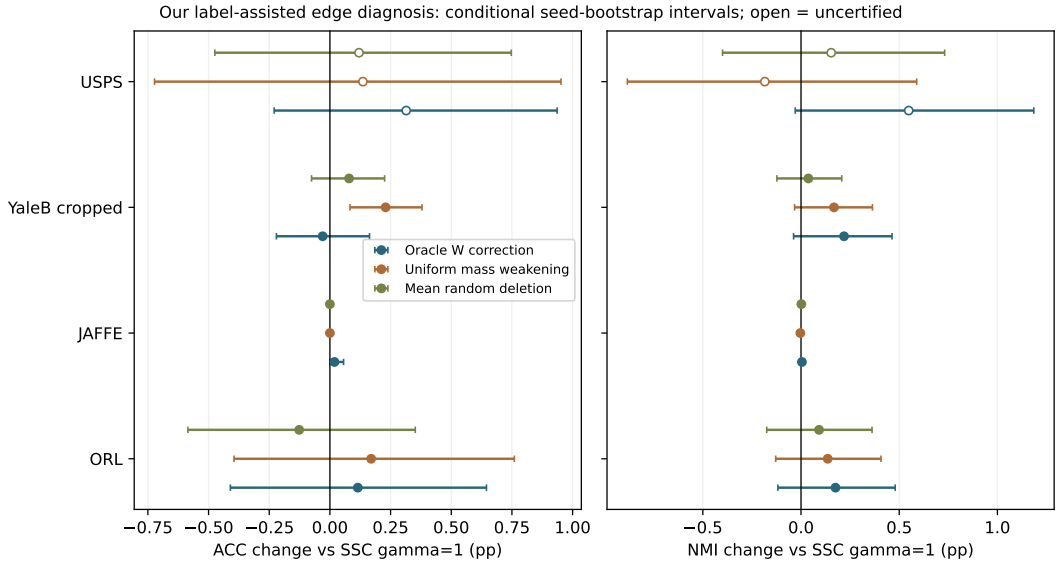


Figure 3: Conditional paired seed intervals at gamma=1. Oracle correction is a label-assisted diagnostic, not an unsupervised method.